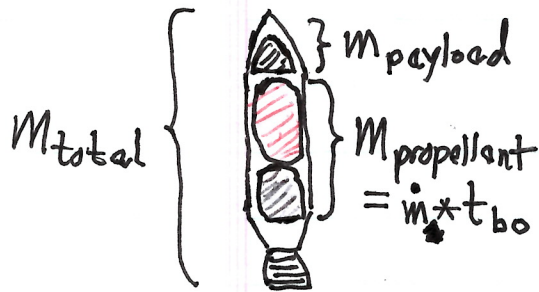


Step 11

⇒ Staged Rocket Mass-over-Time Logic:

> Single stage rocket:



$$M_{empty} = M_{total} - M_{propellant}$$

if $t < t_{burnout}$:

$$M_{tot} = M_{empty} + M_{prop} - \dot{m} * t$$

if $t \geq t_{bo}$:

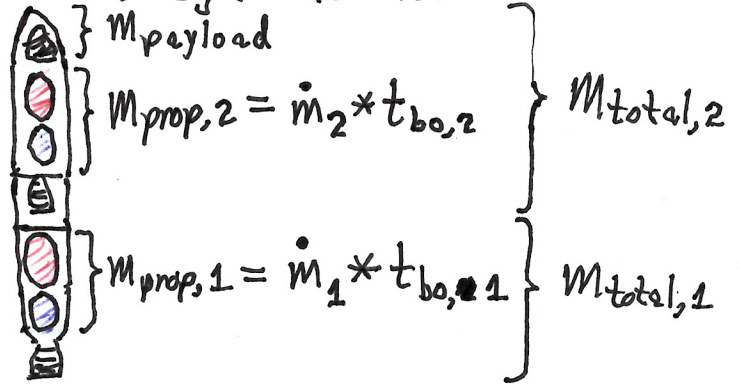
if payload_separates = True:

$$M_{tot} = M_{payload}$$

else:

$$M_{tot} = M_{empty}$$

> Two Stage Rocket



$$M_{empty,1} = M_{total,1} - M_{prop,1}$$

$$M_{empty,2} = M_{total,2} - M_{prop,2}$$

Δt_{coast} = coasting duration after S1 drop & before S2 start



> Two stage rocket mass-over-time logic:

if $t < t_{bo,1}$:

$$M_{total} = M_{total,1} - \dot{m}_1 * t + M_{total,2}$$

if $t \geq t_{bo,1}$ & $t < t_{bo,1} + \Delta t_{coast}$:

$$M_{total} = M_{total,2}$$

if $t \geq t_{bo,2} + \Delta t_{coast}$ & $t < t_{bo,2}$:

$$M_{total} = M_{total,2} - \dot{m}_2 * (t - (t_{bo,2} + \Delta t_{coast}))$$

if $t \geq t_{bo,2}$:

if payload_separates == True:

$$M_{total} = M_{payload}$$

else

$$M_{total} = M_{empty}$$

⇒ Required Rocket Properties

> For whole rocket:

- $M_{payload}$ (float)
- payload_separates (boolean)
- # of stages (integer)
- Δt_{coast}

> For each stage:

- $M_{total,n}$
- $\dot{m}_n$
- $t_{bo,n}$
- $V_{exhaust,n}$
- $M_{prop,n}$
- $M_{empty,n}$
- pitch program

~~if not~~

~~for stage in stages:~~

~~if t >= stage.t_start_abs...~~

~~... and t~~

for i in range(len(stages)):

if t >= stages[i].t_start_abs

⇒ solve for m_other for each rocket stage:

stage 1:

$$m_{\text{other}} = m_{\text{empty2}} + m_{\text{prop2}} \\ + m_{\text{empty3}} + m_{\text{prop3}} \\ + m_{\text{payload}}$$

stage 2:

$$m_{\text{other}} = m_{\text{empty3}} + m_{\text{prop3}} \\ + m_{\text{payload}}$$

Stage 3:

$$m_{\text{other}} = m_{\text{payload}}$$

for i in range(len(stages))

~~range(i)~~ stages[i].m_other = m_payload

for j in range(i+1, len(stages))

$$\text{stages}[i].m_{\text{other}} += \text{stages}[j].m_{\text{empty}} \\ + \text{stages}[j].m_{\text{propellant}}$$

$$\text{stage 1 } m_{\text{other}} = 6000 + 1000 + 1200 \\ = 8200$$

$$\text{stage 2 } m_{\text{other}} = 2200$$